THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE UNIT TEST SET 1

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If $\tan A = \frac{4}{3}$, then $\sin A =$ 1

(a) \frac{4}{3} **(b)** \frac{3}{5} **(c)** \frac{4}{5} **(d)** \frac{5}{4}

2. From a point on level ground, 15 m away from the foot of a vertical tower, the angle of elevation of its top is 60°. The height of the tower is: 1

(a) $30\sqrt{3}$ m **(b)** $15\sqrt{3}$ m **(c)** $5\sqrt{3}$ m **(d)** 15 m

3. If tangents PA and PB from an external point P to a circle with centre O are inclined to each other at 80°, then $\angle POA$ is: 1

(a) 50° **(b)** 100° **(c)** 80° **(d)** 40°

4. A circular field has radius 14 m. Fencing costs ₹ 10 per metre. The cost (in ₹) of fencing the field once is: 1

(a) ₹ 880 **(b)** ₹ 440 **(c)** ₹ 88 **(d)** ₹ 6160

5. If $\cos A = \frac{5}{13}$, then $\tan A =$ 1

(a) \frac{12}{5} **(b)** \frac{12}{13} **(c)** \frac{13}{5} **(d)** \frac{5}{12}

6. A ladder 6 m long leans against a vertical wall and makes an angle of 60° with the level ground. The height reached on the wall is: 1

(a) 3 m **(b)** $\frac{3\sqrt{3}}{2}$ m **(c)** $3\sqrt{3}$ m **(d)** $6\sqrt{3}$ m

7. A point P is 34 cm from the centre O of a circle of radius 16 cm. The length of the tangent drawn from P to the circle is: 1

(a) 50 cm **(b)** 900 cm **(c)** 18 cm **(d)** 30 cm

8. A wheel has radius 21 cm. The number of complete revolutions it makes to cover 66 m is: 1

(a) 100 **(b)** 51 **(c)** 60 **(d)** 50

9. The value of $\sin^2 30^\circ + \cos^2 30^\circ$ is: 1

(a) 0 **(b)** 2 **(c)** 1 **(d)** 3

10. A boy whose eyes are 2 m above the ground stands 10 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 45°. The height of the building is: 1

(a) $12\sqrt{3}$ m **(b)** 13 m **(c)** 10 m **(d)** 12 m

11. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 9$ cm, $BC = 11$ cm and $CD = 7$ cm. The length of AD is: 1

(a) 27 cm **(b)** 13 cm **(c)** 5 cm **(d)** 9 cm

12. The radii of two circles are 19 cm and 9 cm. The radius of the circle whose circumference equals the sum of their circumferences is: 1

(a) 30 cm **(b)** 29 cm **(c)** 56 cm **(d)** 28 cm

13. $\tan 20^\circ \cdot \tan 70^\circ =$ 1

(a) 2 **(b)** 1 **(c)** 20 **(d)** 0

14. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 20 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. The height of the tower is: 1

(a) $20\sqrt{3}$ m **(b)** $10\sqrt{3}$ m **(c)** 10 m **(d)** $\frac{20\sqrt{3}}{3}$ m

15. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 60^\circ$. The $\angle OAB$ is: 1

(a) 30° **(b)** 60° **(c)** 31° **(d)** 29°

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. If $\tan A = \frac{4}{3}$, find $\sec A$. 2

17. A storm breaks a tree at a height of 6 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 30° with the ground. Find the distance from the foot of the tree to the point where its top touches the ground. 2

18. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (3x + 4)$ cm and $QB = (2x + 9)$ cm. Find the length of QA. 2

19. The minute hand of a clock is 14 cm long. Find the distance moved by its tip in 15 minutes. 2

20. Evaluate: $\dfrac{\sin 25^\circ}{\cos 65^\circ} + \dfrac{\tan 40^\circ}{\cot 50^\circ}$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. If $\tan\theta = \dfrac{12}{5}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

22. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 20 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. 3

23. The incircle of $\triangle ABC$ touches BC, CA and AB at D, E and F. $AB = 12$ cm, $BC = 7$ cm and $CA = 9$ cm. Find the lengths AF, BD and CE (in this order). 3

24. A chord of a circle of radius 7 cm subtends a right angle at the centre. Find the area of the minor segment. 3

OR

In $\triangle ABC$, right-angled at B, $AB = 7$ cm and $BC = 24$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

25. From the top of a building 5 m high, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 45°. Find the height of the tower. 3

OR

In $\triangle ABC$, right-angled at B, $AB = 20$ cm and $BC = 21$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Prove that: $\dfrac{\sin\theta}{1+\cos\theta} + \dfrac{1+\cos\theta}{\sin\theta} = 2\csc\theta.$ 5

OR

Two vertical poles of equal height stand on opposite sides of a road 80 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). 5

27. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. 5

OR

A chord of a circle of radius 14 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. 5

END OF QUESTION PAPER Total: 50 Marks

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